WEEK 14 · Bayesian Inference

Bayes and MCMC

Estimates are uncertain too — account for parameter uncertainty

Prior · Likelihood · Posterior · Predictive distribution · Monte Carlo and MCMC · Diagnostics · FFBS

In Week 10 we computed state uncertainty.
Now we also account for not knowing the parameters used in that computation.

Bayes reports parameters as distributions

Three messages that run through this lecture

1

Posterior ∝ likelihood × prior

Report a plausible range instead of a single point, and check whether changing the prior changes the conclusion.

2

Parameter intervals differ from intervals for future outcomes

A forecast adds new shocks on top of parameter estimation error.

3

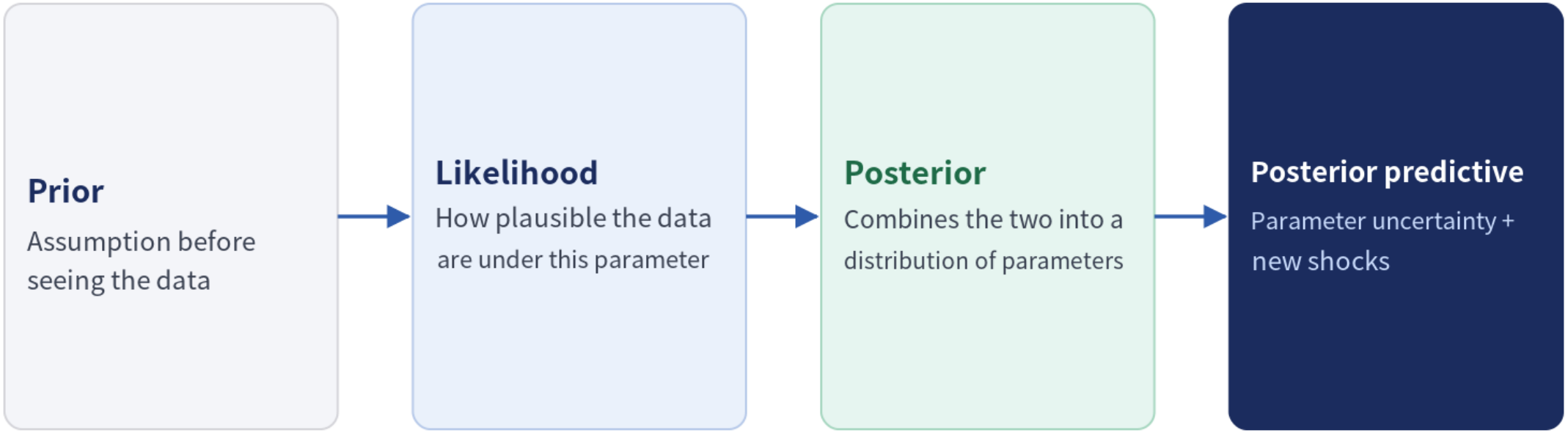
MCMC is a chain that approximates the posterior with samples

Convergence, mixing and effective sample size must be diagnosed; its purpose differs from generating market paths.

The label ‘Bayesian’ does not settle uncertainty — uncertainty conditional on what?

Prior and likelihood make the posterior; the posterior makes the forecast

Combine pre-data assumptions with information in the data to view parameters and the future as distributions



Reporting difference

Reporting beta as a single 1.0 differs from reporting 0.7–1.3 (hypothetical)

01

PART 1 · PRINCIPLES

Prior, Likelihood, Posterior, Prediction

Bayes' theorem · Credible intervals · Posterior predictive distribution

The posterior is proportional to the likelihood times the prior

θ is the parameter, Y the observed data

$$\underbrace{p(\theta \mid Y)}_{\text{posterior}} \propto \underbrace{p(Y \mid \theta)}_{\text{likelihood}} \underbrace{p(\theta)}_{\text{prior}}$$

Three elements

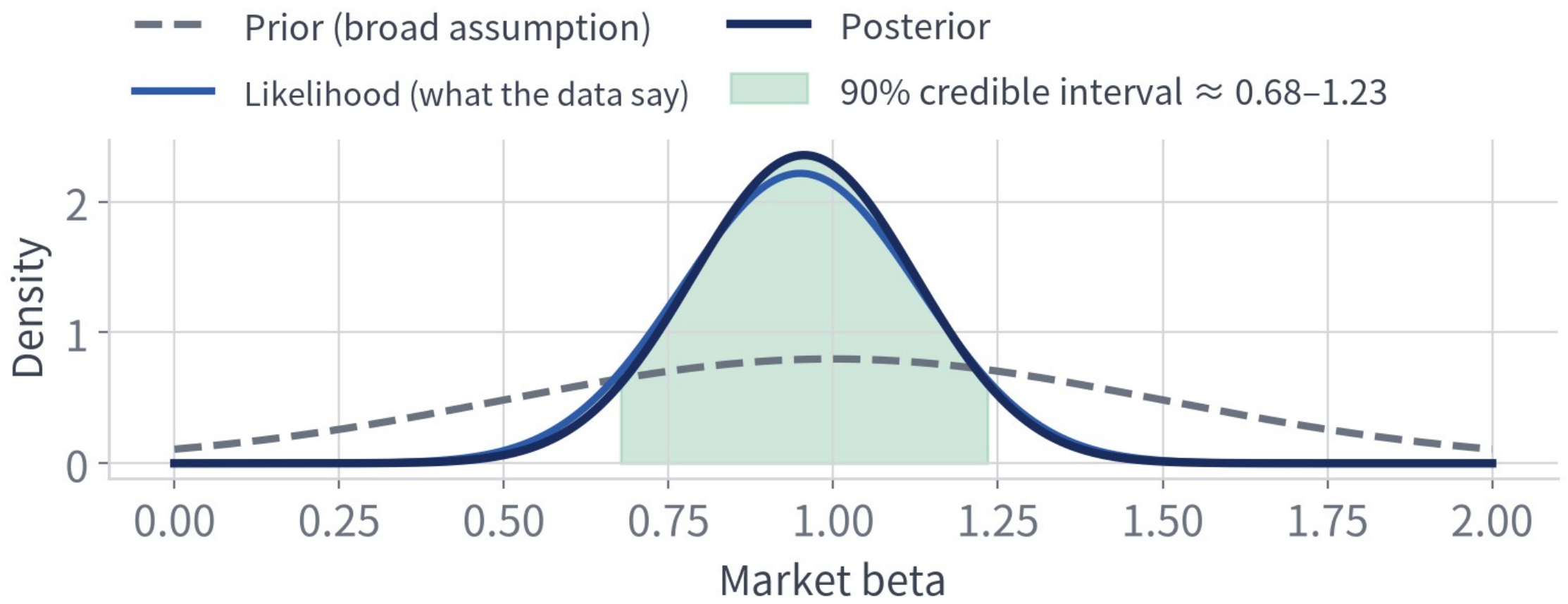
- Prior: the assumption before seeing data
- Likelihood: how plausible the data are under this parameter
- Posterior: the result of combining the two

Sensitivity check

- Check whether the conclusion changes when the prior changes

With more data, the posterior narrows toward the likelihood

Prior, likelihood and posterior of a market beta drawn as normal distributions · ※ illustrative values



A forecast averages each parameter's future distribution over the posterior

Posterior predictive distribution

$$\underbrace{p(y_{\text{new}} \mid Y)}_{\text{posterior predictive}} = \int \underbrace{p(y_{\text{new}} \mid \theta)}_{\text{future distribution for each } \theta} \underbrace{p(\theta \mid Y)}_{\text{posterior}} \, d\theta$$

Two uncertainties

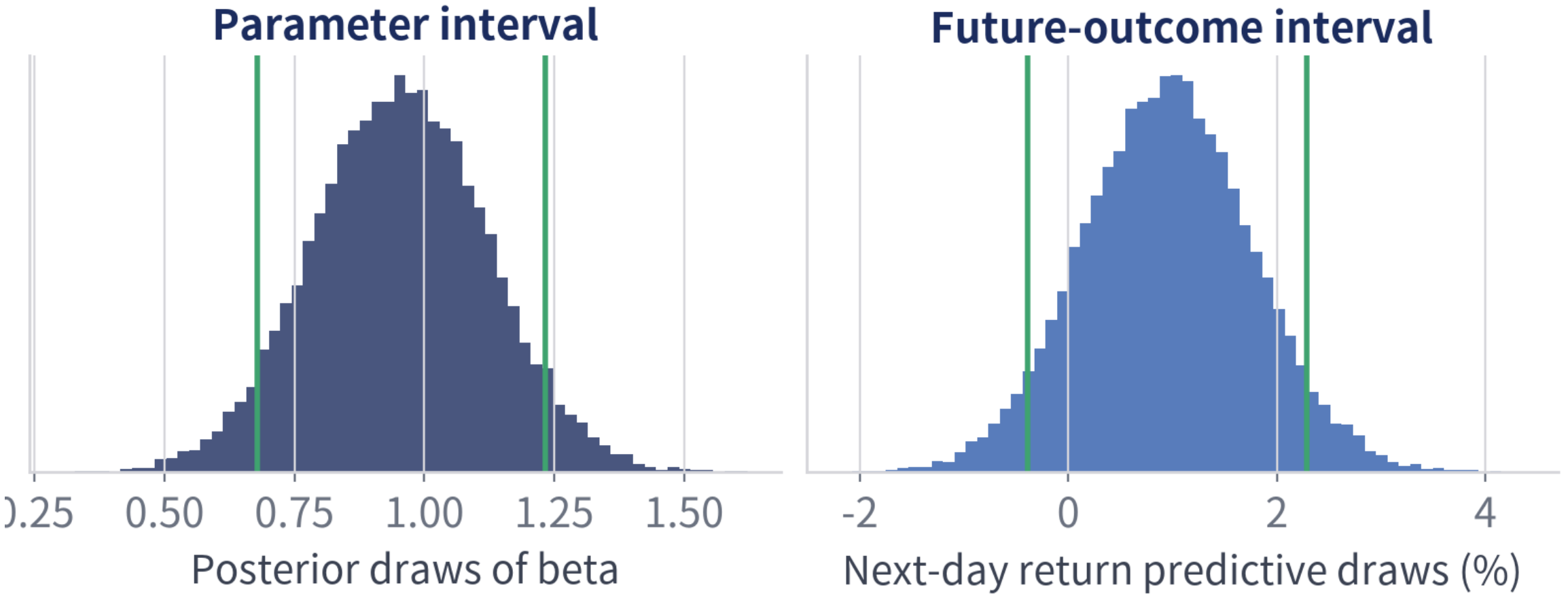
- Parameter estimation error
- New future shocks

Do not confuse

- Credible interval for beta ≠ prediction interval for next-day return

The interval for future outcomes is wider than the parameter interval

Left: posterior draws of beta · Right: next-day returns from those betas plus new shocks · ※ illustrative values



02

PART 2 · COMPUTATION

Monte Carlo and MCMC

Source of samples · Samplers · Diagnostics

Three sampling computations differ in where the samples come from

MCMC does not generate financial price paths, and iteration numbers are not market dates

Independent Monte Carlo

Draws independently and directly from the target distribution
Check: sample size · standard error

MCMC

Chain with the target posterior as its stationary distribution
Check: convergence · mixing · effective sample size

Week 16 market simulation

Generates future paths from an assumed market model
Check: distributional fit · path dependence · risk assessment

Each sampler moves differently and has its own pitfalls

Metropolis–Hastings · Gibbs · HMC·NUTS

Method	Intuition	Watch out for
Metropolis–Hastings	Propose a candidate, accept it with some probability	Balance of moves vs rejections set by proposal width
Gibbs	Draw one block at a time with the others fixed	Slow movement when blocks are strongly correlated
HMC·NUTS	Use gradients to move efficiently through the posterior	Divergences, bad geometry, reparameterization

Diagnostics

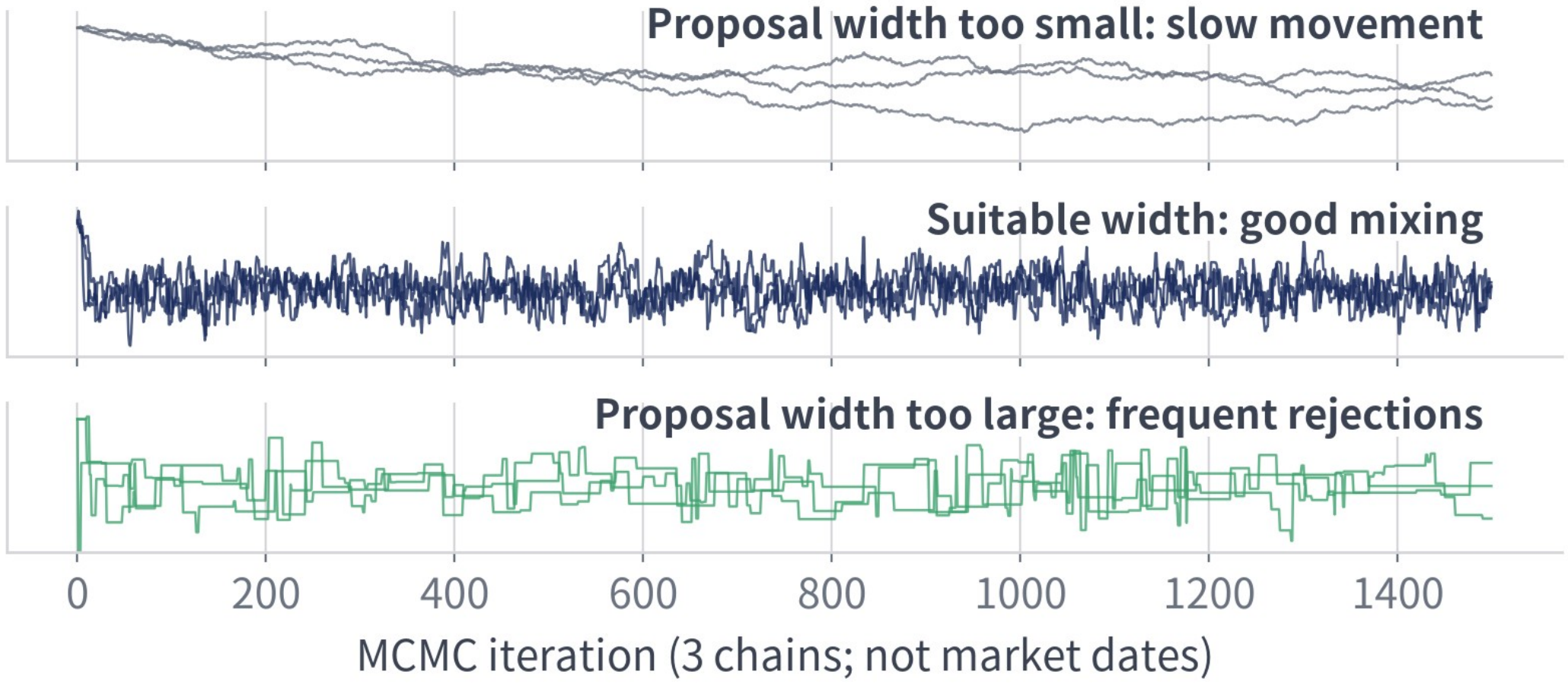
- Multiple chains · warm-up · trace plots
- Effective sample size · R-hat · divergences

Limits

One good diagnostic does not guarantee model fit — also run posterior predictive checks

The chain mixes well only with a suitable proposal width

Same target sampled by three Metropolis–Hastings chains · starting points deliberately far off · ※ illustrative



03

PART 3 · CONNECTIONS

Two Layers: Kalman and Bayes

States and parameters · FFBS · Lab

State uncertainty splits into two parts

Law of total variance — compare within the same information set

$$\mathrm{Var}(x_t \mid Y) = \underbrace{E[\mathrm{Var}(x_t \mid Y, \theta) \mid Y]}_{\text{left even if parameters are known}} + \underbrace{\mathrm{Var}(E[x_t \mid Y, \theta] \mid Y)}_{\text{due to differing parameters}}$$

First term

- State uncertainty that remains even with known parameters — the Week 10 filter variance

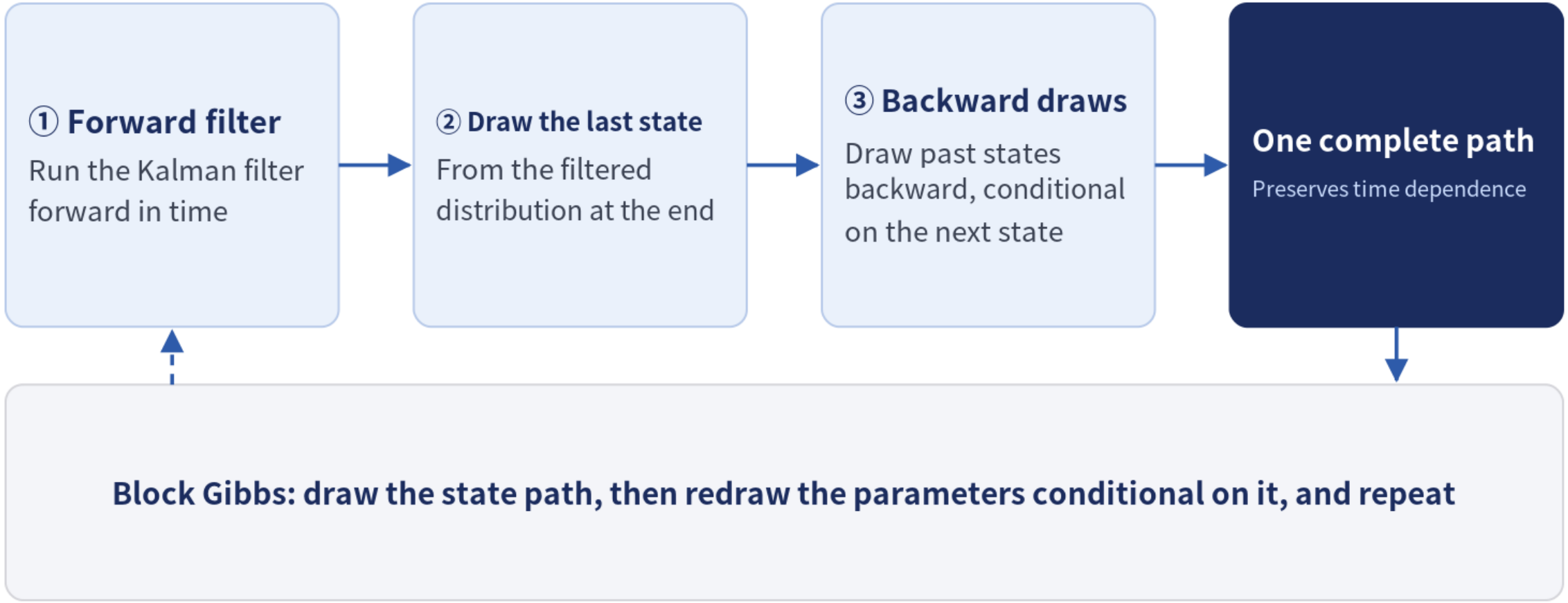
Second term

- Variation in state estimates caused by differing parameters

This does not mean a Bayesian interval is always wider than any point-estimate interval

FFBS draws one complete path of the states

Conditionally linear Gaussian model · filtering and smoothing summarize the distribution; FFBS draws a path from it



The lab starts by reporting the uncertainty of beta

Paste into Claude Code · dynamic beta and FFBS are optional extensions

[Input] Prompt · Claude Code

```
Generate a beta regression from synthetic asset and market returns
and fit a Bayesian regression.
Record the seed and the true beta, and
compare the OLS estimate and interval with the posterior median and credible
interval.
Plot the prediction interval for future returns and the beta interval
in separate figures.
Run MCMC with multiple chains and diagnostics,
and report the effective sample size.
Show prior sensitivity and posterior predictive checks.
Implement dynamic beta and FFBS only as an optional extension
if time permits.
Distinguish future dates in a financial simulation
from MCMC iteration numbers.
```

Check

- Beta and prediction intervals plotted separately?
- R-hat, effective sample size, divergences
- Does the conclusion change with the prior?

Report estimates as distributions too

Next week: forecast ranges with deep learning and foundation models

1

Posterior ∝ likelihood × prior

Report a range instead of a point estimate and check prior sensitivity.

2

Parameter interval ≠ prediction interval

Future outcomes carry both estimation error and new shocks.

3

Use MCMC with diagnostics

Multiple chains, R-hat, effective sample size, posterior predictive checks — iterations are not dates.

Week 15 forecast ranges · Week 16 generated paths: which uncertainty, conditional on what?